Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/05/2006
PI Name:	Chan, Moses H.W. Ph.D.		
Project Title:	Critical Casimir Forces		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	16802	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	04/01/2001	**End Date:**	02/28/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Garcia, Rafael (Worcester Polytechnic Institute)		
Grant/Contract No.:	NAG 3 2891		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		
Task Description:	To task objective is being accomplished (1) by constructing new sample cell with silicon wafers as electrodes to measure the thinning of helium film as the cell is cooled through the superfluid transition (2)by constructing a Casimir force apparatus consist of a flexible diaphragm separated at a distance of a micorn from a fixed electrode, allowing the measurement of the Casimir force directly (3) testing the device at cryogenic temperatures. The objective is to measure fluctuation-induced forces in the vicinity of critical points of three different universality class: the liquid gas critical point, the lambda point in He-4 and the He-3-He-4 Mixture Tricritical point. We want to make the measuremen on a film that is adsorbed on a well surface such as silicon thatis atomically flat. We have made measurement on the change in the chemical potential as the system is brought through the critical point, we want in the seond experiment to measure the force directly. One of the most beautiful aspects of physics is how pheonomena in widely different systems are described by the same mathematical formulation. Directly analogous to the Casimir force in electromagnetism, the critical Casimir		

	force is predicted to be a universal feature of all critical points, containing important information about finite size effects and boundary conditions. The result of this work will provide the first measurement of the critical Casimir effect near a liquid-gas critical point and it will allow us to examine the effect of varying boundary condition and different universality class on the properties of fluids near their critical points.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This experiment will help to bring a thorough understanding of the nature of Casimir force and how it is analogous to the quantum Casimir effect. Student and post-doc working on this project receive rigorous training both in theory and in many different techniques in experimental physics.
Task Progress:	We have completed a set of careful measurements on critcial Casmir force induced thinning of adsorbed heliums films near the superfluid transition. the films were adsorbed on 'virgin' silicon wafers assembed in ultra-clean enviroment. In an earlier paper, Garcia and Chan (Phys. Rev. Lett. 83, 1187 (1999)) reported the observation of the effect of helium films adsorbed on copper surfaces. Casimir effect induced thinning of the films were seen. However, the results show systematic deviation from the predictions of finite size scaling theory, a theory thought to be exact. Indeed theories were proposed to explain the deviation. A source of uncertainty is the far from ideal condition of the copper surface. It is rough with scratches and dust particles. In contrast the surface of silicon is excedingly smooth and clean. The new results on three different helium films showed that finite size scaling theory works perfectly in predicitng the position and amplitude of the amount of film thinning. This is an important result in that while the finite size scaling is a sophisticated and theory widely accepted by the theoretical physics community, experimental confirmation is exceedingly sparse. The quantitative confirmation from this experiment is an important milestone on the subject. The result of this experiment has been submitted for publication to Physical Review Letters.
Bibliography Type:	Description: (Last Updated: 09/11/2009)
Articles in Peer-reviewed Journals	Ganshin A, Scheidemantel S, Garcia R, Chan MH. "Critical Casimir force in 4He films: confirmation of finite-size scaling." Phys Rev Lett. 2006 Aug 18;97(7):075301. PubMed PMID: 17026241 , Aug-2006